the higher temperature metal to form a metal hydride/metal carbonate, which produces heat used to generate electricity," he added.

Professor Chris Moran, deputy vice chancellor research, Curtin University, said, "As with lithium battery systems that Curtin is also developing, deployment of a cost-effective energy storage system using thermal batteries will revolutionise the landscape of renewable energy production worldwide by allowing renewables to truly compete with fossil fuels.

"While a lithium battery stores electrical energy that can be used to provide electricity when the sun is not shining, this thermal battery stores heat from concentrated solar thermal, which can be used when the sun is not shining to run a turbine to produce electricity."

Semi-liquid metal anode for next-generation batteries

Researchers from Carnegie Mellon University's Mellon College of Science and College of Engineering have developed a semi-liquid lithium metal-based anode that represents a new standard in battery design. Lithium batteries made using this new electrode type could have a higher capacity and be much safer than typical lithium metal-based batteries that use lithium foil as anode.

Lithium-based batteries are one of the most common types of rechargeable battery used in modern electronics due to their ability to store high amounts of energy. Traditionally, these batteries are made of combustible liquid electrolytes and two electrodes, an anode and a cathode, which are separated by a membrane. After a battery has been charged and discharged repeatedly, strands of lithium called dendrites can grow on the surface of the electrode. The dendrites can pierce through the membrane that separates the two electrodes. This allows contact between the anode and cathode, which can cause the battery to short circuit and, in the worst case, catch fire.

One proposed solution to the volatile liquid electrolytes used in current batteries is to replace these with solid ceramic electrolytes. These electrolytes are highly conductive, non-combustible and strong enough to resist dendrites. However, researchers have found that contact between the ceramic electrolyte and a solid lithium anode is insufficient for storing and supplying the amount of power needed for most electronics.

Sipei Li, a doctoral student in Carnegie Mellon's Department of Chemistry, and Han Wang, a doctoral student in Carnegie Mellon's Department of Materials Science and Engineering, were able to overcome this shortcoming by creating a new class of material that can be used as a semi-liquid metal anode.

Li and Wang created a dual-conductive polymer/carbon composite matrix that has lithium microparticles evenly distributed throughout. The matrix remains flowable at room temperatures, which allows it to create a sufficient level of contact with the solid electrolyte. By combining the semi-liquid metal anode with a garnet-based solid ceramic electrolyte, they were able to cycle the cell at ten times higher current density than cells with a solid electrolyte and a traditional lithium foil anode. This cell also had a much longer cycle-life than traditional cells.

Robots can now relieve chronic pain

Based on an analysis of the patient by a thermal camera, a collaborative robot applies targeted laser therapy to pain hotspots (Credit: Swinburne University of Technology)

Researchers at Swinburne University have developed a collaborative robot system to automatically treat back, neck and head pain caused by soft tissue injury. Based on an analysis of the patient by a thermal camera, the system uses a collaborative robot to apply targeted laser therapy to identified pain hotspots.

Senior lecturer at Swinburne University, Dr Mats Isaksson, who led the project, said, "Unlike conventional industrial robots that operate in a cage, collaborative robots are designed to work alongside people. They are power- and speed-limited so they can collide with people without causing harm."

In collaboration with industry partner IR Robotics, the focus of Dr Isaksson's project was to develop an automatic solution for safe robotic examination and treatment of patients with chronic pain.

Dr Isaksson added, "Using the same technology used in cricket to show whether the ball has made contact with the bat, a thermal camera scans the patient, and locates injuries and inflammation by identifying hotspots in a thermal image.

"Location of the hotspot is then sent to the collaborative robot that moves a low-level infrared (IR) laser into contact with the patient to perform treatment."